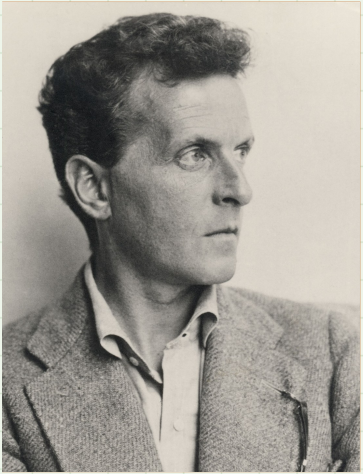
Advanced Machine Learning, ETH Zurich

Attention

How a word reads the words around it

Carlos Cotrini

Autumn semester 2026

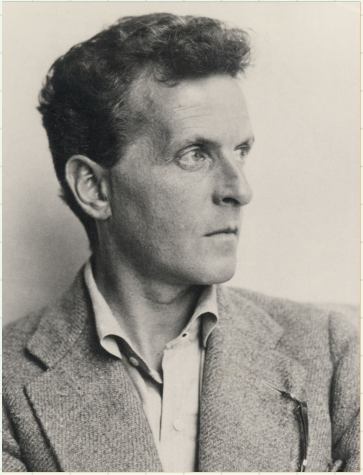


Moritz Nähr, 1930
Ludwig Wittgenstein

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1889 to 1951

Born in Vienna. He studied engineering in Berlin and Manchester, then logic with Bertrand Russell at Cambridge, where he later taught philosophy.



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The pages that follow quote his *Philosophische Untersuchungen*, Philosophical Investigations, published in 1953, two years after his death, and *The Blue Book* and *The Brown Book*, which he dictated in English at Cambridge between 1933 and 1935.

Subject

§ 1: Augustine

Entry

When they [my elders] named any thing, and as they spoke turned towards it, I saw and remembered that they called what they would point out by the name they uttered. And that they meant this thing and no other was plain from the motion of their body, the natural language, as it were, of all nations, expressed by the countenance, glances of the eye, gestures of the limbs, and tones of the voice, indicating the affections of the mind, as it pursues, possesses, rejects, or shuns. And thus by constantly hearing words, as they occurred in various sentences, I collected gradually for what they stood; and having broken in my mouth to these signs, I thereby gave utterance to my will.

Augustine, *Confessions* I.8, in the English of E. B. Pusey (1838). *Philosophische Untersuchungen* §1 opens with this passage in Latin, with the insertion *majores homines*, rendered here in brackets.

Subject

§ 1: a picture of language

Entry

In diesen Worten erhalten wir, so scheint es mir, ein bestimmtes Bild von dem Wesen der menschlichen Sprache. Nämlich dieses: Die Wörter der Sprache benennen Gegenstände; Sätze sind Verbindungen von solchen Benennungen.

In diesem Bild von der Sprache finden wir die Wurzeln der Idee: Jedes Wort hat eine Bedeutung. Diese Bedeutung ist dem Wort zugeordnet. Sie ist der Gegenstand, für welchen das Wort steht.

Ludwig Wittgenstein, *Philosophische Untersuchungen* (1953), § 1. English translation for this lecture.

Subject

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In these words we get, so it seems to me, a particular picture of the essence of human language. Namely this: the words of language name objects; sentences are combinations of such names.

In this picture of language we find the roots of the idea: every word has a meaning. This meaning is assigned to the word. It is the object for which the word stands.

Ludwig Wittgenstein, *Philosophische Untersuchungen* (1953), § 1. English translation for this lecture.

Subject

§ 1: kinds of words

Entry

Von einem Unterschied der Wortarten spricht Augustinus nicht. Wer das Lernen der Sprache so beschreibt, denkt, so möchte ich glauben, zunächst an Hauptwörter, wie »Tisch«, »Stuhl«, »Brot«, und die Namen von Personen, erst in zweiter Linie an die Namen gewisser Tätigkeiten und Eigenschaften, und an die übrigen Wortarten als etwas, was sich finden wird.

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Augustine does not speak of a difference between kinds of words. Whoever describes the learning of language in this way thinks, so I would like to believe, first of nouns, such as "table", "chair", "bread", and of the names of persons, only secondarily of the names of certain activities and properties, and of the remaining kinds of words as something that will sort itself out.

Ludwig Wittgenstein, *Philosophische Untersuchungen* (1953), § 1. English translation for this lecture.

Subject

§ 1: five red apples

Entry

Denke nun an diese Verwendung der Sprache: Ich schicke jemand einkaufen. Ich gebe ihm einen Zettel, auf diesem stehen die Zeichen: »fünf rote Äpfel«. Er trägt den Zettel zum Kaufmann; der öffnet die Lade, auf welcher das Zeichen »Äpfel« steht; dann sucht er in einer Tabelle das Wort »rot« auf und findet ihm gegenüber ein Farbmuster; nun sagt er die Reihe der Grundzahlwörter (ich nehme an, er weiß sie auswendig) bis zum Worte »fünf« und bei jedem Zahlwort nimmt er einen Apfel aus der Lade, der die Farbe des Musters hat. So, und ähnlich, operiert man mit Worten.

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Now think of this use of language: I send someone shopping. I give him a slip of paper, on which stand the signs: “five red apples”. He carries the slip to the shopkeeper; the latter opens the drawer on which the sign “apples” stands; then he looks up the word “red” in a table and finds a colour sample opposite it; now he says the series of the cardinal numerals (I assume that he knows them by heart) up to the word “five”, and at each numeral he takes from the drawer an apple that has the colour of the sample.

So, and similarly, one operates with words.

Ludwig Wittgenstein, *Philosophische Untersuchungen* (1953), § 1. English translation for this lecture.

Subject

§ 1: the word »fünf«

Entry

»Wie weiß er aber, wo und wie er das Wort »rot« nachschlagen soll und was er mit dem Wort »fünf« anzufangen hat?«

Nun, ich nehme an, er handelt, wie ich es beschrieben habe. Die Erklärungen haben irgendwo ein Ende.

Was ist aber die Bedeutung des Wortes »fünf«?

Von einer solchen war hier gar nicht die Rede; nur davon, wie das Wort »fünf« gebraucht wird.

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“But how does he know where and how he is to look up the word ‘red’ and what he is to do with the word ‘five’?”

Well, I assume that he acts as I have described. Explanations come to an end somewhere.

But what is the meaning of the word “five”?

There was no talk at all of such a thing here, only of how the word “five” is used.

Ludwig Wittgenstein, *Philosophische Untersuchungen* (1953), § 1. English translation for this lecture.

Subject

The Blue Book: a red flower

Entry

[. . .] If I give someone the order "fetch me a red flower from that meadow", how is he to know what sort of flower to bring, as I have only given him a word?

Now the answer one might suggest first is that he went to look for a red flower carrying a red image in his mind, and comparing it with the flowers to see which of them had the colour of the image. Now there is such a way of searching, and it is not at all essential that the image we use should be a mental one. In fact the process may be this:

I carry a chart co-ordinating names and coloured squares. When I hear the order "fetch me etc." I draw my finger across the chart from the word "red" to a certain square, and I go and look for a flower which has the same colour as the square. But this is not the only way of searching and it isn't the usual way. We go, look about us, walk up to a flower and pick it, without comparing it to anything. [. . .]

Ludwig Wittgenstein, *The Blue Book*, dictated in English at Cambridge in 1933 and 1934.

Subject

The Brown Book: five slabs

Entry

[. . .] The builder's man knows by heart the series of words from one to ten. On being given the order, "Five slabs!", he goes to where the slabs are kept, says the words from one to five, takes up a plate for each word, & carries them to the builder. [. . .] Learning the numerals by heart will be one of the essential features of learning this language. [. . .]

(Remark: [. . .] by introducing numerals we have introduced an entirely different kind of instrument into our language. The difference of kind is much more obvious when we contemplate such a simple example than when we look at our ordinary language with innumerable kinds of words all looking more or less alike when they stand in the dictionary. [. . .])

Ludwig Wittgenstein, *The Brown Book*, dictated in English at Cambridge in 1934 and 1935, Part I, 2).

Subject

The Brown Book: five apples

Entry

[. . .] consider this language-game: A sends B to various houses in their town to fetch goods of various sorts from various people. A gives B various lists. On top of every list he puts a scribble, and B is trained to go to that house on the door of which he finds the same scribble, this is the name of the house. In the first column of every list he then finds one or more scribbles which he has been taught to read out. When he enters the house he calls out these words, and every inhabitant of the house has been trained to run up to him when a certain one of these sounds is called out, these sounds are the names of the people. He then addresses himself to each one of them in turn and shews to each two consecutive scribbles which stand on the list against his name. The first of these two, people of that town have been trained to associate with some particular kind of object, say, apples. The second is one of a row of scribbles which each man carries about him on a slip of paper. The person thus addressed fetches, say, five apples. The first scribble was the generic name of the objects required, the second, the name of their number.

Ludwig Wittgenstein, *The Brown Book*, dictated in English at Cambridge in 1934 and 1935, Part II.

Subject

§ 43: the word »Bedeutung«

Entry

Man kann für eine große Klasse von Fällen der Benützung des Wortes »Bedeutung« (wenn auch nicht für alle Fälle seiner Benützung) dieses Wort so erklären: Die Bedeutung eines Wortes ist sein Gebrauch in der Sprache.

Und die Bedeutung eines Namens erklärt man manchmal dadurch, daß man auf seinen Träger zeigt.

Ludwig Wittgenstein, *Philosophische Untersuchungen* (1953), § 43. English translation for this lecture.

Subject

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Und die Bedeutung eines Namens erklärt man manchmal dadurch, daß man auf seinen Träger zeigt.

For a large class of cases of the use of the word “meaning” (though not for all cases of its use) one can explain this word thus: the meaning of a word is its use in the language.

And the meaning of a name is sometimes explained by pointing to its bearer.

Ludwig Wittgenstein, *Philosophische Untersuchungen* (1953), § 43. English translation for this lecture.

Subject

§ 558: the word »ist«

Entry

Was heißt es, daß im Satze »Die Rose ist rot« das »ist« eine andere Bedeutung hat als in »zwei mal zwei ist vier«? Wenn man antwortet, es heiße, daß verschiedene Regeln von diesen beiden Wörtern gelten, so ist zu sagen, daß wir hier nur ein Wort haben.

Und wenn ich nur auf die grammatischen Regeln achte, so erlauben diese eben die Verwendung des Wortes »ist« in beiderlei Zusammenhängen.

Die Regel aber, welche zeigt, daß das Wort »ist« in diesen Sätzen verschiedene Bedeutung hat, ist die, welche erlaubt, im zweiten Satz das Wort »ist« durch das Gleichheitszeichen zu ersetzen, und die diese Ersetzung im ersten Satz verbietet.

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What does it mean that in the sentence “The rose is red” the “is” has a different meaning from that in “two times two is four”? If one answers that it means that different rules hold for these two words, then it must be said that we have only one word here.

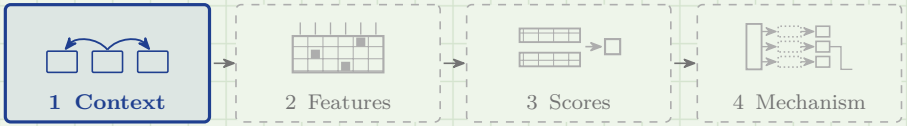
And if I attend only to the grammatical rules, these do allow the use of the word “is” in both kinds of context.

But the rule which shows that the word “is” has different meanings in these sentences is the one which allows the word “is” in the second sentence to be replaced by the equals sign, and which forbids this replacement in the first.

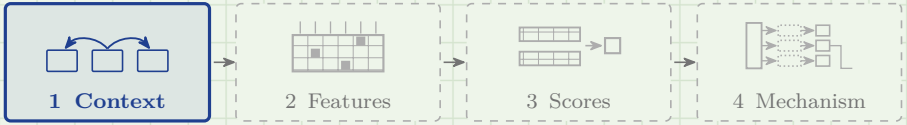
Ludwig Wittgenstein, *Philosophische Untersuchungen* (1953), § 558. English translation for this lecture.

Entry 1

Words in context



The question. In the language model, every row of E holds one word, yet the next word depends on the whole prefix. So row i must read the rows above it. How?



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Now. Attention: every row reads the other rows and adds to itself what it finds. In Wuggish, the toy language of this lecture, every word finds what it needs in the rows above it.

Subject

A word takes its meaning from the words around it

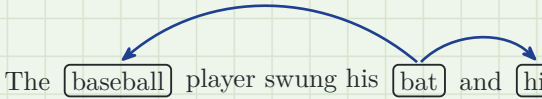
Entry 1

2

The same word can mean two different things. The words around it decide which one.

The baseball player swung his bat and hit a home run.

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Here **bat** reads baseball and hit: a club.

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The baseball player swung his bat and hit a home run.

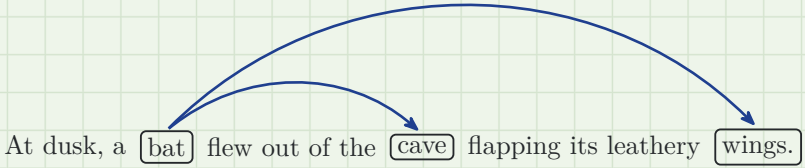
Here **bat** reads baseball and hit: a club.

At dusk, a bat flew out of the cave flapping its leathery wings.

The same word can mean two different things. The words around it decide which one.



Here **bat** reads baseball and hit: a club.



Here **bat** reads cave and wings: an animal.

Few people know the words alacrity and torpor. Read the sentence: its other words tell you what they mean.

She responded with **alacrity**, leaping to her feet with energy,

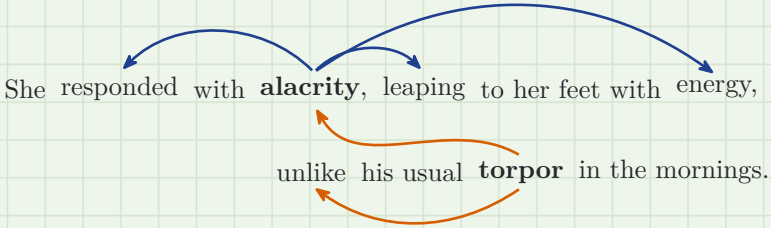
unlike his usual **torpor** in the mornings.

Few people know the words alacrity and torpor. Read the sentence: its other words tell you what they mean.



alacrity reads responded, leaping and energy: something quick and lively.

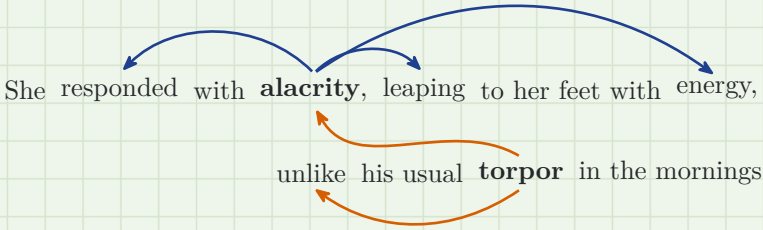
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torpor reads unlike and alacrity: the opposite of alacrity.

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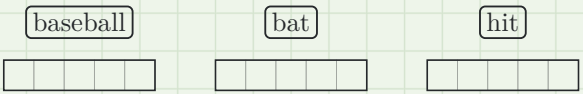
alacrity: brisk and cheerful readiness.



torpor: a state of physical or mental inactivity.

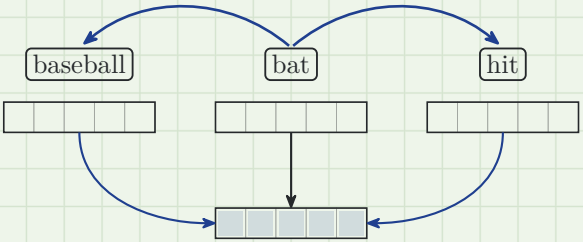
A network that reads a sentence does it in two steps, and every word does both.

1. Every word gets a representation: a row of numbers.



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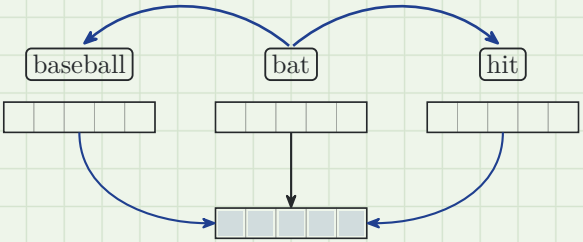


bat, now the club

2. Every word reads the other words and adds to its row what it finds.

A network that reads a sentence does it in two steps, and every word does both.

1. Every word gets a representation: a row of numbers.



bat, now the club

2. Every word reads the other words and adds to its row what it finds.

This is *attention*. The rest of the lecture computes it by hand, on a language small enough to see every number.

Entry 2

Wuggish



So far. A word takes its meaning from the words around it. Attention lets every word read the other words and add what it finds to its own row.



So far. A word takes its meaning from the words around it. Attention lets every word read the other words and add what it finds to its own row.

Now. A toy language of eleven words, Wuggish, and a way to write each word as numbers: a row of seven features, each a 1 or a 0.

Wuggish is a made-up sister language of English. Its sentences describe a cat, a duck and a fish, each plain or with a look: running, in boots, swimming or diving.

kat



dukk



feesh



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		runnin	*bootet*
kat			
dukk			
feesh			

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kat					
dukk					
feesh					

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		runnin	*bootet*	*swimmin*	*divin*
kat			 ***pussinbootz***		
dukk		 ***runnerdukk***		 ***pondukk***	
feesh					 ***divefeesh***

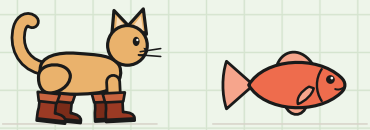
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		runnin	*bootet*	*swimmin*	*divin*
kat			 ***pussinbootz***		
dukk		 ***runnerdukk***		 ***pondukk***	
feesh					 ***divefeesh***

Eleven words: three things, four looks, and four things with a look. *pussinbootz* is the cat in boots that *bootet kat* draws.

Learn the eleven words from pictures, then read 12 sentences whose words come in a fresh order every time. At the end, the game shows what you have been doing all along.

bootet *feesh* *kat*



Play it now, on a laptop or a phone: scan the code, or open the address below.

eth-aml-hs26.github.io/neural-networks/viz/wuggish-game

To compute with words, we ask every word the same seven questions. Each question is a *feature*, and its answer is 1 for yes and 0 for no.

What a thing has

LEGS has legs? 

FINS has fins? 

WINGS has wings? 

The nouns say yes here: a cat has legs, a duck legs and wings, a fish fins.

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An adjective brings its look, and so does a word that names a thing with a look.

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Where they meet

earth

water

A cat runs and wears boots on earth, a fish swims and dives in water, and a duck does all four.

Earth and water get no column of their own.

What a look brings

RUNNING is running? 

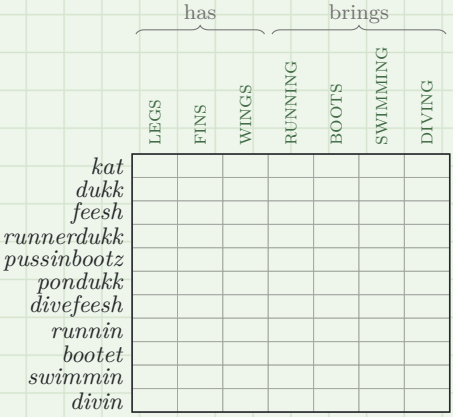
BOOTS wears boots? 

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Ask every Wuggish word the seven questions and write its answers in a row, its *embedding*. The eleven rows form the *embedding table*.



11 words by $d = 7$ features

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	has			brings			
	LEGS	FINS	WINGS	RUNNING	BOOTS	SWIMMING	DIVING
kat	**1**	0	0	0	0	0	0
dukk	**1**	0	**1**	0	0	0	0
feesh	0	**1**	0	0	0	0	0
runnerdukk							
pussinbootz							
pondukk							
divefeesh							
runnin							
bootet							
swimmin							
divin							

11 words by $d = 7$ features

kat, *dukk* and *feesh* say what they have: legs; legs and wings; fins.

Ask every Wuggish word the seven questions and write its answers in a row, its *embedding*. The eleven rows form the *embedding table*.

	has			brings			
	LEGS	FINS	WINGS	RUNNING	BOOTS	SWIMMING	DIVING
kat	**1**	0	0	0	0	0	0
dukk	**1**	0	**1**	0	0	0	0
feesh	0	**1**	0	0	0	0	0
runnerdukk	**1**	0	**1**	**1**	0	0	0
pussinbootz	**1**	0	0	0	**1**	0	0
pondukk	**1**	0	**1**	0	0	**1**	0
divefeesh	0	**1**	0	0	0	0	**1**
runnin							
bootet							
swimmin							
divin							

11 words by $d = 7$ features

kat, *dukk* and *feesh* say what they have: legs; legs and wings; fins.

runnerdukk, *pussinbootz*, *pondukk* and *divefeesh* each add one look: Puss in Boots is a cat in boots.

Ask every Wuggish word the seven questions and write its answers in a row, its *embedding*. The eleven rows form the *embedding table*.

	has			brings			
	LEGS	FINS	WINGS	RUNNING	BOOTS	SWIMMING	DIVING
kat	**1**	0	0	0	0	0	0
dukk	**1**	0	**1**	0	0	0	0
feesh	0	**1**	0	0	0	0	0
runnerdukk	**1**	0	**1**	**1**	0	0	0
pussinbootz	**1**	0	0	0	**1**	0	0
pondukk	**1**	0	**1**	0	0	**1**	0
divefeesh	0	**1**	0	0	0	0	**1**
runnin	0	0	0	**1**	0	0	0
bootet	0	0	0	0	**1**	0	0
swimmin	0	0	0	0	0	**1**	0
divin	0	0	0	0	0	0	**1**

11 words by $d = 7$ features

kat, *dukk* and *feesh* say what they have: legs; legs and wings; fins.

runnerdukk, *pussinbootz*, *pondukk* and *divefeesh* each add one look: Puss in Boots is a cat in boots.

An adjective says what its look brings, with a single 1.

Ask every Wuggish word the seven questions and write its answers in a row, its *embedding*. The eleven rows form the *embedding table*.

	has			brings			
	LEGS	FINS	WINGS	RUNNING	BOOTS	SWIMMING	DIVING
kat	**1**	0	0	0	0	0	0
dukk	**1**	0	**1**	0	0	0	0
feesh	0	**1**	0	0	0	0	0
runnerdukk	**1**	0	**1**	**1**	0	0	0
pussinbootz	**1**	0	0	0	**1**	0	0
pondukk	**1**	0	**1**	0	0	**1**	0
divefeesh	0	**1**	0	0	0	0	**1**
runnin	0	0	0	**1**	0	0	0
bootet	0	0	0	0	**1**	0	0
swimmin	0	0	0	0	0	**1**	0
divin	0	0	0	0	0	0	**1**

11 words by $d = 7$ features

kat, *dukk* and *feesh* say what they have: legs; legs and wings; fins.

runnerdukk, *pussinbootz*, *pondukk* and *divefeesh* each add one look: Puss in Boots is a cat in boots.

An adjective says what its look brings, with a single 1.

We wrote these rows by hand, to read them. A trained network learns its own rows, and its features have no names.

A word of a sentence, a *token*, is a row of the table. A feature is a column: a question that several words answer with yes.

	has			brings			
	LEGS	FINS	WINGS	RUNNING	BOOTS	SWIMMING	DIVING
kat	**1**	0	0	0	0	0	0
dukk	**1**	0	**1**	0	0	0	0
feesh	0	**1**	0	0	0	0	0
runnerdukk	**1**	0	**1**	**1**	0	0	0
pussinbootz	**1**	0	0	0	**1**	0	0
pondukk	**1**	0	**1**	0	0	**1**	0
divefeesh	0	**1**	0	0	0	0	**1**
runnin	0	0	0	**1**	0	0	0
bootet	0	0	0	0	**1**	0	0
swimmin	0	0	0	0	0	**1**	0
divin	0	0	0	0	0	0	**1**

A word of a sentence, a *token*, is a row of the table. A feature is a column: a question that several words answer with yes.

	has			brings			
	LEGS	FINS	WINGS	RUNNING	BOOTS	SWIMMING	DIVING
kat	**1**	0	0	0	0	0	0
dukk	**1**	0	**1**	0	0	0	0
feesh	0	**1**	0	0	0	0	0
runnerdukk	**1**	0	**1**	**1**	0	0	0
pussinbootz	**1**	0	0	0	**1**	0	0
pondukk	**1**	0	**1**	0	0	**1**	0
divefeesh	0	**1**	0	0	0	0	**1**
runnin	0	0	0	**1**	0	0	0
bootet	0	0	0	0	**1**	0	0
swimmin	0	0	0	0	0	**1**	0
divin	0	0	0	0	0	0	**1**

A token: the row of *pussinbootz*, seven numbers that say a cat in boots.

A word of a sentence, a *token*, is a row of the table. A feature is a column: a question that several words answer with yes.

	has			brings			
	LEGS	FINS	WINGS	RUNNING	BOOTS	SWIMMING	DIVING
kat	**1**	0	0	0	0	0	0
dukk	**1**	0	**1**	0	0	0	0
feesh	0	**1**	0	0	0	0	0
runnerdukk	**1**	0	**1**	**1**	0	0	0
pussinbootz	**1**	0	0	0	**1**	0	0
pondukk	**1**	0	**1**	0	0	**1**	0
divefeesh	0	**1**	0	0	0	0	**1**
runnin	0	0	0	**1**	0	0	0
bootet	0	0	0	0	**1**	0	0
swimmin	0	0	0	0	0	**1**	0
divin	0	0	0	0	0	0	**1**

A token: the row of *pussinbootz*, seven numbers that say a cat in boots.

A feature: the column LEGS. Five words say yes, *kat*, *dukk*, *runnerdukk*, *pussinbootz* and *pondukk*: legs are what they share.

A word of a sentence, a *token*, is a row of the table. A feature is a column: a question that several words answer with yes.

	has			brings			
	LEGS	FINS	WINGS	RUNNING	BOOTS	SWIMMING	DIVING
kat	**1**	0	0	0	0	0	0
dukk	**1**	0	**1**	0	0	0	0
feesh	0	**1**	0	0	0	0	0
runnerdukk	**1**	0	**1**	**1**	0	0	0
pussinbootz	**1**	0	0	0	**1**	0	0
pondukk	**1**	0	**1**	0	0	**1**	0
divefeesh	0	**1**	0	0	0	0	**1**
runnin	0	0	0	**1**	0	0	0
bootet	0	0	0	0	**1**	0	0
swimmin	0	0	0	0	0	**1**	0
divin	0	0	0	0	0	0	**1**
words	**5**	**2**	**3**	**2**	**2**	**2**	**2**

A token: the row of *pussinbootz*, seven numbers that say a cat in boots.

A feature: the column LEGS. Five words say yes, *kat*, *dukk*, *runnerdukk*, *pussinbootz* and *pondukk*: legs are what they share.

Every column holds the 1s of two words or more: the count under it.

A word of a sentence, a *token*, is a row of the table. A feature is a column: a question that several words answer with yes.

	has			brings			
	LEGS	FINS	WINGS	RUNNING	BOOTS	SWIMMING	DIVING
kat	**1**	0	0	0	0	0	0
dukk	**1**	0	**1**	0	0	0	0
feesh	0	**1**	0	0	0	0	0
runnerdukk	**1**	0	**1**	**1**	0	0	0
pussinbootz	**1**	0	0	0	**1**	0	0
pondukk	**1**	0	**1**	0	0	**1**	0
divefeesh	0	**1**	0	0	0	0	**1**
runnin	0	0	0	**1**	0	0	0
bootet	0	0	0	0	**1**	0	0
swimmin	0	0	0	0	0	**1**	0
divin	0	0	0	0	0	0	**1**
words	**5**	**2**	**3**	**2**	**2**	**2**	**2**

A token: the row of *pussinbootz*, seven numbers that say a cat in boots.

A feature: the column LEGS. Five words say yes, *kat*, *dukk*, *runnerdukk*, *pussinbootz* and *pondukk*: legs are what they share.

Every column holds the 1s of two words or more: the count under it.

And there are more words than features: 11 rows, 7 columns. So a feature can never be a second name for a word.

Two rows of the table add up to a third. The row of *pussinbootz* holds the row of *kat* and one more 1, at BOOTS: the row of *bootet*.

	LEGS	FINS	WINGS	RUNNING	BOOTS	SWIMMING	DIVING
row of *kat*	**1**	0	0	0	0	0	0



Two rows of the table add up to a third. The row of *pussinbootz* holds the row of *kat* and one more 1, at BOOTS: the row of *bootet*.

	LEGS	FINS	WINGS	RUNNING	BOOTS	SWIMMING	DIVING
row of *kat*	**1**	0	0	0	0	0	0
+ row of *bootet*	0	0	0	0	**1**	0	0



Two rows of the table add up to a third. The row of *pussinbootz* holds the row of *kat* and one more 1, at BOOTS: the row of *bootet*.

	LEGS	FINS	WINGS	RUNNING	BOOTS	SWIMMING	DIVING
row of *kat*	1	0	0	0	0	0	0
+ row of *bootet*	0	0	0	0	1	0	0
row of *pussinbootz*	1	0	0	0	1	0	0



Two rows of the table add up to a third. The row of *pussinbootz* holds the row of *kat* and one more 1, at BOOTS: the row of *bootet*.

	LEGS	FINS	WINGS	RUNNING	BOOTS	SWIMMING	DIVING
row of *kat*	1	0	0	0	0	0	0
+ row of *bootet*	0	0	0	0	1	0	0
row of *pussinbootz*	1	0	0	0	1	0	0



The other three alike: *runnerdukk* is *dukk* plus *runnin*, *pondukk* is *dukk* plus *swimmin*, and *divefeesh* is *feesh* plus *divin*.

Two rows of the table add up to a third. The row of *pussinbootz* holds the row of *kat* and one more 1, at BOOTS: the row of *bootet*.

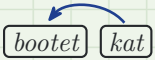
	LEGS	FINS	WINGS	RUNNING	BOOTS	SWIMMING	DIVING
row of *kat*	1	0	0	0	0	0	0
+ row of *bootet*	0	0	0	0	1	0	0
row of *pussinbootz*	1	0	0	0	1	0	0



The other three alike: *runnerdukk* is *dukk* plus *runnin*, *pondukk* is *dukk* plus *swimmin*, and *divefeesh* is *feesh* plus *divin*.

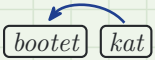
Keep this sum in mind: in our sentence, attention will add the row of *bootet* to the row of *kat*.

Wuggish has one rule of order: an adjective stands somewhere before the noun it describes. Other words may stand between them, and beyond that the order carries nothing.

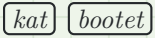


✓ a cat in boots

Wuggish has one rule of order: an adjective stands somewhere before the noun it describes. Other words may stand between them, and beyond that the order carries nothing.

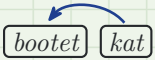


✓ a cat in boots

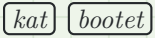


✗ the adjective stands after its noun

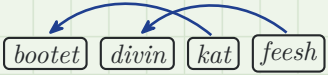
Wuggish has one rule of order: an adjective stands somewhere before the noun it describes. Other words may stand between them, and beyond that the order carries nothing.



✓ a cat in boots

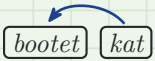


✗ the adjective stands after its noun

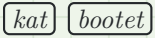


✓ other words between them

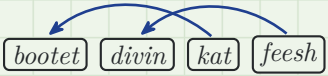
Wuggish has one rule of order: an adjective stands somewhere before the noun it describes. Other words may stand between them, and beyond that the order carries nothing.



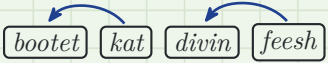
✓ a cat in boots



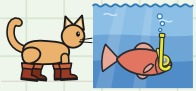
✗ the adjective stands after its noun



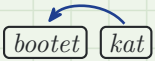
✓ other words between them



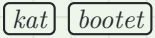
✓ the sentence we compute



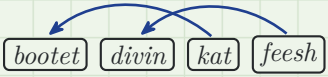
Wuggish has one rule of order: an adjective stands somewhere before the noun it describes. Other words may stand between them, and beyond that the order carries nothing.



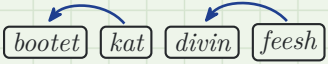
✓ a cat in boots



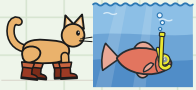
✗ the adjective stands after its noun



✓ other words between them



✓ the sentence we compute



So a noun finds its adjectives among the words before it: in E, in the rows above its own.

Write our sentence one word per row and replace every word by its row of the table. The result is E, with $n = 4$ rows and $d = 7$ columns.

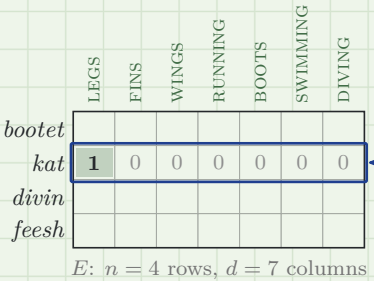
	LEGS	FINS	WINGS	RUNNING	BOOTS	SWIMMING	DIVING
bootet							
kat							
divin							
feesh							

E: $n = 4$ rows, $d = 7$ columns

	LEGS	FINS	WINGS	RUNNING	BOOTS	SWIMMING	DIVING
kat	**1**	0	0	0	0	0	0
dukk	**1**	0	**1**	0	0	0	0
feesh	0	**1**	0	0	0	0	0
runnerdukk	**1**	0	**1**	**1**	0	0	0
pussinbootz	**1**	0	0	0	**1**	0	0
pondukk	**1**	0	**1**	0	0	**1**	0
divefeesh	0	**1**	0	0	0	0	**1**
runnin	0	0	0	**1**	0	0	0
bootet	0	0	0	0	**1**	0	0
swimmin	0	0	0	0	0	**1**	0
divin	0	0	0	0	0	0	**1**

the embedding table

Write our sentence one word per row and replace every word by its row of the table. The result is E, with $n = 4$ rows and $d = 7$ columns.



E: $n = 4$ rows, $d = 7$ columns

	LEGS	FINS	WINGS	RUNNING	BOOTS	SWIMMING	DIVING
kat	**1**	0	0	0	0	0	0
dukk	**1**	0	**1**	0	0	0	0
feesh	0	**1**	0	0	0	0	0
runnerdukk	**1**	0	**1**	**1**	0	0	0
pussinbootz	**1**	0	0	0	**1**	0	0
pondukk	**1**	0	**1**	0	0	**1**	0
divefeesh	0	**1**	0	0	0	0	**1**
runnin	0	0	0	**1**	0	0	0
bootet	0	0	0	0	**1**	0	0
swimmin	0	0	0	0	0	**1**	0
divin	0	0	0	0	0	0	**1**

the embedding table

Write our sentence one word per row and replace every word by its row of the table. The result is E, with $n = 4$ rows and $d = 7$ columns.

	LEGS	FINS	WINGS	RUNNING	BOOTS	SWIMMING	DIVING
bootet	0	0	0	0	**1**	0	0
kat	**1**	0	0	0	0	0	0
divin	0	0	0	0	0	0	**1**
feesh	0	**1**	0	0	0	0	0

E: $n = 4$ rows, $d = 7$ columns

	LEGS	FINS	WINGS	RUNNING	BOOTS	SWIMMING	DIVING
kat	**1**	0	0	0	0	0	0
dukk	**1**	0	**1**	0	0	0	0
feesh	0	**1**	0	0	0	0	0
runnerdukk	**1**	0	**1**	**1**	0	0	0
pussinbootz	**1**	0	0	0	**1**	0	0
pondukk	**1**	0	**1**	0	0	**1**	0
divefeesh	0	**1**	0	0	0	0	**1**
runnin	0	0	0	**1**	0	0	0
bootet	0	0	0	0	**1**	0	0
swimmin	0	0	0	0	0	**1**	0
divin	0	0	0	0	0	0	**1**

the embedding table

Write our sentence one word per row and replace every word by its row of the table. The result is E, with $n = 4$ rows and $d = 7$ columns.

	LEGS	FINS	WINGS	RUNNING	BOOTS	SWIMMING	DIVING
bootet	0	0	0	0	**1**	0	0
kat	**1**	0	0	0	0	0	0
divin	0	0	0	0	0	0	**1**
feesh	0	**1**	0	0	0	0	0

E: $n = 4$ rows, $d = 7$ columns

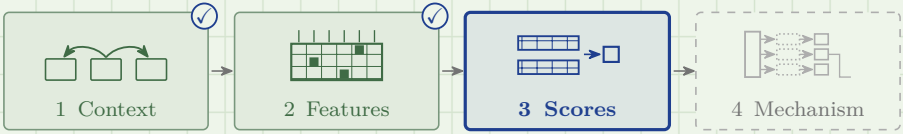
The row of *kat* holds LEGS alone: a cat. Nothing in it says booted, and *bootet* stands in the row above. Attention will make the row of *kat* read the row of *bootet*.

	LEGS	FINS	WINGS	RUNNING	BOOTS	SWIMMING	DIVING
kat	**1**	0	0	0	0	0	0
dukk	**1**	0	**1**	0	0	0	0
feesh	0	**1**	0	0	0	0	0
runnerdukk	**1**	0	**1**	**1**	0	0	0
pussinbootz	**1**	0	0	0	**1**	0	0
pondukk	**1**	0	**1**	0	0	**1**	0
divefeesh	0	**1**	0	0	0	0	**1**
runnin	0	0	0	**1**	0	0	0
bootet	0	0	0	0	**1**	0	0
swimmin	0	0	0	0	0	**1**	0
divin	0	0	0	0	0	0	**1**

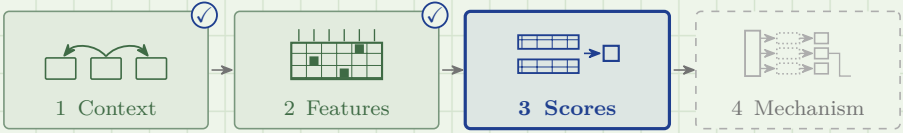
the embedding table

Entry 3

Queries, keys and scores



So far. Every word is a row of seven features, and our sentence is a block E. In E, the row of *kat* says nothing about boots.



So far. Every word is a row of seven features, and our sentence is a block E. In E, the row of *kat* says nothing about boots.

Now. Every word must find the rows to add to its own. We recall the game, picture it as a shopping mall, and let queries and keys find the rows.

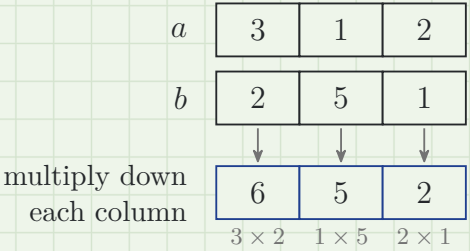
Two rows of the same length give one number, their *inner product*.

a	3	1	2

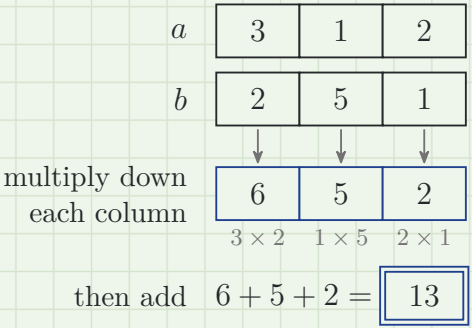
Two rows of the same length give one number, their *inner product*.

a	3	1	2
b	2	5	1

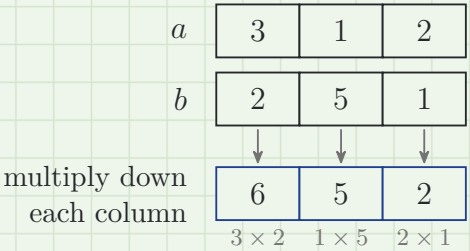
Two rows of the same length give one number, their *inner product*.



Two rows of the same length give one number, their *inner product*.



Two rows of the same length give one number, their *inner product*.

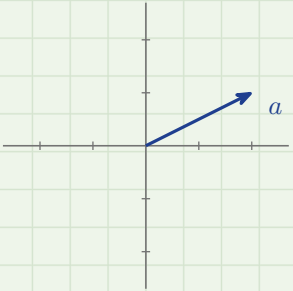


then add $6 + 5 + 2 = \boxed{13}$ the inner product of a and b

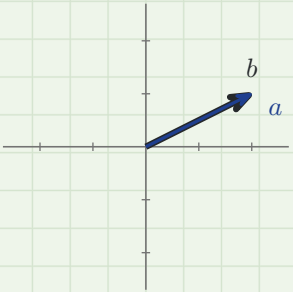
inner product of a and $b = a_1b_1 + a_2b_2 + a_3b_3$

Read a row of two numbers as an arrow. Keep a and turn b: the inner product falls as the angle between them opens.

$a = (2, 1)$



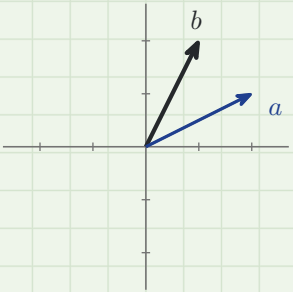
Read a row of two numbers as an arrow. Keep a and turn b: the inner product falls as the angle between them opens.



$a = (2, 1)$

$b = (2, 1)$ $\quad 2 \times 2 + 1 \times 1 = 5$ same direction

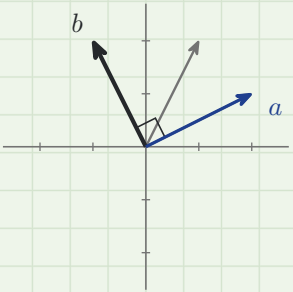
Read a row of two numbers as an arrow. Keep a and turn b: the inner product falls as the angle between them opens.



$a = (2, 1)$

$b = (2, 1)$	$2 \times 2 + 1 \times 1 = 5$	same direction
$b = (1, 2)$	$2 \times 1 + 1 \times 2 = 4$	close directions

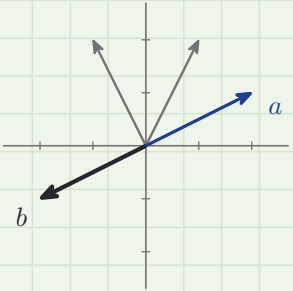
Read a row of two numbers as an arrow. Keep a and turn b: the inner product falls as the angle between them opens.



$a = (2, 1)$

$b = (2, 1)$	$2 \times 2 + 1 \times 1 = 5$	same direction
$b = (1, 2)$	$2 \times 1 + 1 \times 2 = 4$	close directions
$b = (-1, 2)$	$2 \times (-1) + 1 \times 2 = 0$	at right angles

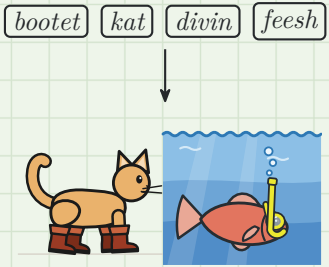
Read a row of two numbers as an arrow. Keep a and turn b: the inner product falls as the angle between them opens.



$a = (2, 1)$

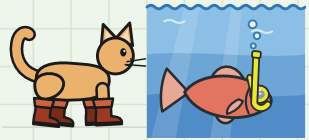
$b = (2, 1)$	$2 \times 2 + 1 \times 1 = 5$	same direction
$b = (1, 2)$	$2 \times 1 + 1 \times 2 = 4$	close directions
$b = (-1, 2)$	$2 \times (-1) + 1 \times 2 = 0$	at right angles
$b = (-2, -1)$	$2 \times (-2) + 1 \times (-1) = -5$	opposite directions

Think back to the game. Our sentence *bootet kat divin feesh* draws a cat in boots and a fish that dives. How did you find that picture?



Think back to the game. Our sentence *bootet kat divin feesh* draws a cat in boots and a fish that dives. How did you find that picture?

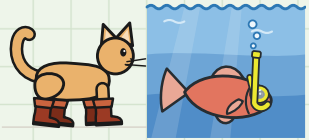
bootet *kat* *divin* *feesh*



1 Take the next animal of the sentence.

Think back to the game. Our sentence *bootet kat divin feesh* draws a cat in boots and a fish that dives. How did you find that picture?

bootet *kat* *divin* *feesh*

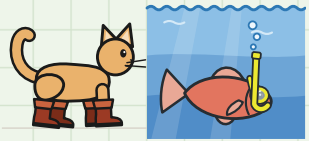


1 Take the next animal of the sentence.

2 Collect the adjectives that fit it.

Think back to the game. Our sentence *bootet kat divin feesh* draws a cat in boots and a fish that dives. How did you find that picture?

bootet *kat* *divin* *feesh*



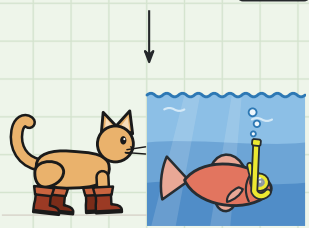
1 Take the next animal of the sentence.

2 Collect the adjectives that fit it.

3 Give it their looks: add their rows to its own row.

Think back to the game. Our sentence *bootet kat divin feesh* draws a cat in boots and a fish that dives. How did you find that picture?

bootet *kat* *divin* *feesh*



next animal

1 Take the next animal of the sentence.

2 Collect the adjectives that fit it.

3 Give it their looks: add their rows to its own row.

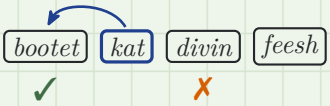
Our sentence has two animals, so you went round the loop twice: first for *kat*, then for *feesh*.

Round one of the loop. The first animal of our sentence is *kat*: a cat, which walks on earth.

1 Take the next animal: *kat*.

bootet *kat* *divin* *feesh*

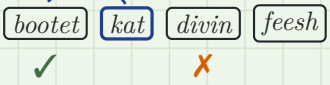
Round one of the loop. The first animal of our sentence is *kat*: a cat, which walks on earth.



1 Take the next animal: *kat*.

2 Collect the adjectives that fit it. *bootet* fits: boots go on earth. *divin* stays out: diving happens in water.

Round one of the loop. The first animal of our sentence is *kat*: a cat, which walks on earth.



	LEGS	FINS	WINGS	RUNNING	BOOTS	SWIMMING	DIVING
kat	**1**	0	0	0	0	0	0
+ *bootet*	0	0	0	0	**1**	0	0

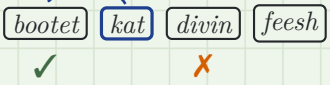


1 Take the next animal: *kat*.

2 Collect the adjectives that fit it. *bootet* fits: boots go on earth. *divin* stays out: diving happens in water.

3 Add their rows to its own. *kat* gains BOOTS: the row of *pussinbootz*, the cat in boots.

Round one of the loop. The first animal of our sentence is *kat*: a cat, which walks on earth.



	LEGS	FINS	WINGS	RUNNING	BOOTS	SWIMMING	DIVING
kat	1	0	0	0	0	0	0
+ *bootet*	0	0	0	0	1	0	0
= *pussinbootz*	1	0	0	0	1	0	0



1 Take the next animal: *kat*.

2 Collect the adjectives that fit it. *bootet* fits: boots go on earth. *divin* stays out: diving happens in water.

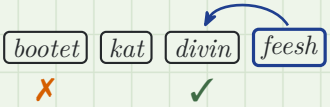
3 Add their rows to its own. *kat* gains BOOTS: the row of *pussinbootz*, the cat in boots.

Round two of the loop. The next animal is *feesh*: a fish, which swims in water.

bootet *kat* *divin* *feesh*

1 Take the next animal: *feesh*.

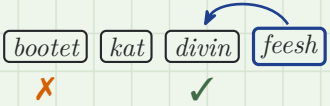
Round two of the loop. The next animal is *feesh*: a fish, which swims in water.



1 Take the next animal: *feesh*.

2 Collect the adjectives that fit it.
divin fits: diving happens in water.
bootet stays out: boots go on earth.

Round two of the loop. The next animal is *feesh*: a fish, which swims in water.



	LEGS	FINS	WINGS	RUNNING	BOOTS	SWIMMING	DIVING
feesh	0	**1**	0	0	0	0	0
+ *divin*	0	0	0	0	0	0	**1**

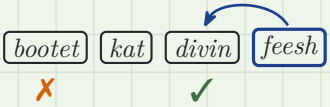


1 Take the next animal: *feesh*.

2 Collect the adjectives that fit it. *divin* fits: diving happens in water. *bootet* stays out: boots go on earth.

3 Add their rows to its own. *feesh* gains DIVING: the row of *divefeesh*, the fish that dives.

Round two of the loop. The next animal is *feesh*: a fish, which swims in water.



	LEGS	FINS	WINGS	RUNNING	BOOTS	SWIMMING	DIVING
feesh	0	**1**	0	0	0	0	0
+ *divin*	0	0	0	0	0	0	**1**
= *divefeesh*	0	**1**	0	0	0	0	**1**

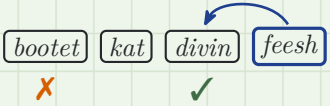


1 Take the next animal: *feesh*.

2 Collect the adjectives that fit it. *divin* fits: diving happens in water. *bootet* stays out: boots go on earth.

3 Add their rows to its own. *feesh* gains DIVING: the row of *divefeesh*, the fish that dives.

Round two of the loop. The next animal is *feesh*: a fish, which swims in water.



	LEGS	FINS	WINGS	RUNNING	BOOTS	SWIMMING	DIVING
feesh	0	**1**	0	0	0	0	0
+ *divin*	0	0	0	0	0	0	**1**
= *divefeesh*	0	**1**	0	0	0	0	**1**



1 Take the next animal: *feesh*.

2 Collect the adjectives that fit it. *divin* fits: diving happens in water. *bootet* stays out: boots go on earth.

3 Add their rows to its own. *feesh* gains DIVING: the row of *divefeesh*, the fish that dives.

No animal is left: the loop ends, and the sentence is drawn.

A network sees no pictures and knows no rules of a game. It has only the numbers of E, and from them alone it must find, for every word, the rows to add to its own.

	LEGS	FINS	WINGS	RUNNING	BOOTS	SWIMMING	DIVING
bootet	0	0	0	0	**1**	0	0
kat	**1**	0	0	0	0	0	0
divin	0	0	0	0	0	0	**1**
feesh	0	**1**	0	0	0	0	0

E

A network sees no pictures and knows no rules of a game. It has only the numbers of E, and from them alone it must find, for every word, the rows to add to its own.

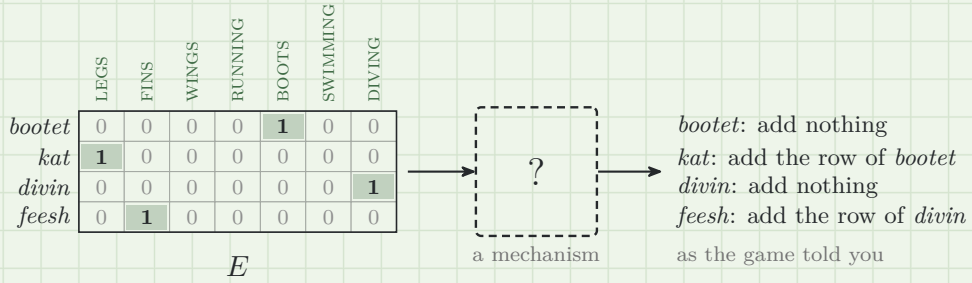
	LEGS	FINS	WINGS	RUNNING	BOOTS	SWIMMING	DIVING
bootet	0	0	0	0	**1**	0	0
kat	**1**	0	0	0	0	0	0
divin	0	0	0	0	0	0	**1**
feesh	0	**1**	0	0	0	0	0

E

bootet: add nothing
kat: add the row of *bootet*
divin: add nothing
feesh: add the row of *divin*

as the game told you

A network sees no pictures and knows no rules of a game. It has only the numbers of E, and from them alone it must find, for every word, the rows to add to its own.



In the game the pictures told you which looks fit. A network needs a mechanism that reads the same answers off the numbers of E, for any sentence.

To find that mechanism, picture a sentence as a shopping mall. Its nouns go shopping, and its adjectives sell.



Every **noun** is a shopper with a cart, and its own animal is in the cart already: *kat* comes with a plain cat.

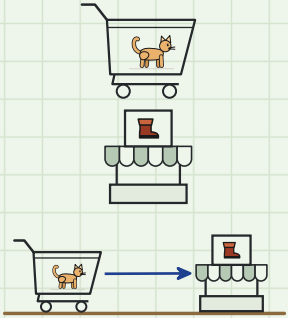
To find that mechanism, picture a sentence as a shopping mall. Its nouns go shopping, and its adjectives sell.



Every **noun** is a shopper with a cart, and its own animal is in the cart already: *kat* comes with a plain cat.

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To find that mechanism, picture a sentence as a shopping mall. Its nouns go shopping, and its adjectives sell.

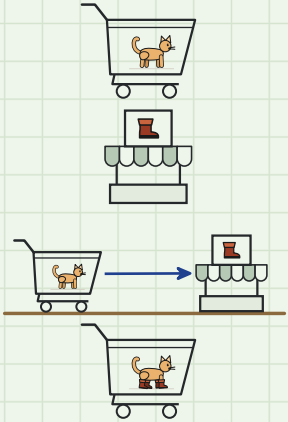


Every **noun** is a shopper with a cart, and its own animal is in the cart already: *kat* comes with a plain cat.

Every **adjective** is a stand, with a sign that shows what it sells: *bootet* sells boots.

A shopper buys at the stands that sell what it looks for. A cat shops on earth, and a fish shops in water.

To find that mechanism, picture a sentence as a shopping mall. Its nouns go shopping, and its adjectives sell.



Every **noun** is a shopper with a cart, and its own animal is in the cart already: *kat* comes with a plain cat.

Every **adjective** is a stand, with a sign that shows what it sells: *bootet* sells boots.

A shopper buys at the stands that sell what it looks for. A cat shops on earth, and a fish shops in water.

At the exit, the cart holds the new row: *kat* leaves as a cat in boots.

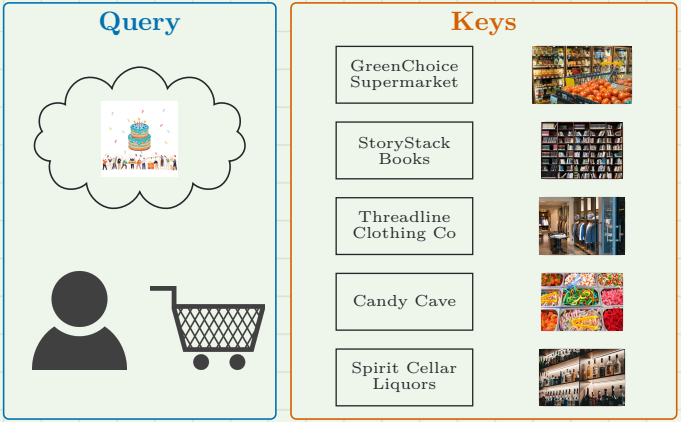
You go shopping for a party. What you look for is the *query*; the signs over the shops are the *keys*; what you carry home are the *values*.

Query





You go shopping for a party. What you look for is the *query*; the signs over the shops are the *keys*; what you carry home are the *values*.



You go shopping for a party. What you look for is the *query*; the signs over the shops are the *keys*; what you carry home are the *values*.

Query





Keys

GreenChoice Supermarket



StoryStack Books



Threadline Clothing Co



Candy Cave



Spirit Cellar Liquors



Values













You go shopping for a party. What you look for is the *query*; the signs over the shops are the *keys*; what you carry home are the *values*.

Query



Keys

GreenChoice Supermarket



StoryStack Books



Threadline Clothing Co



Candy Cave



Spirit Cellar Liquors



Values







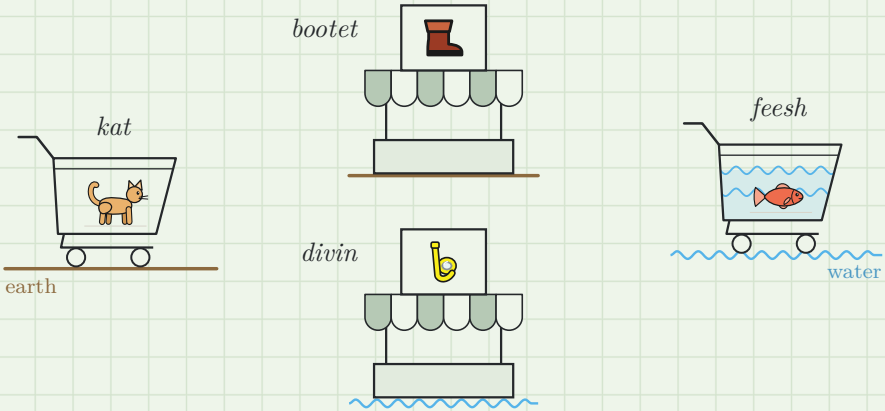




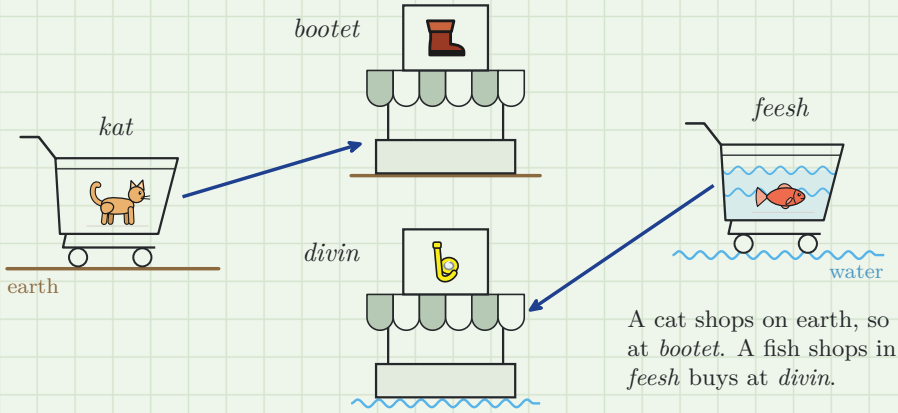


The party matches three keys. Their values go home.

Our sentence makes a small mall: two shoppers, *kat* and *feesh*, and two stands, *bootet* and *divin*.

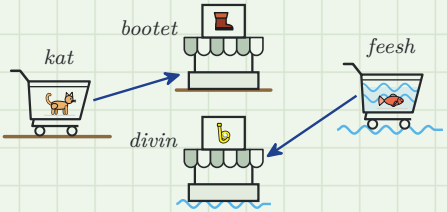


Our sentence makes a small mall: two shoppers, *kat* and *feesh*, and two stands, *bootet* and *divin*.



A cat shops on earth, so *kat* buys at *bootet*. A fish shops in water, so *feesh* buys at *divin*.

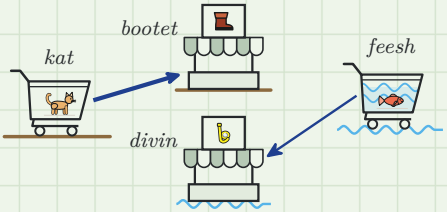
Write the arrows into a table S, one row and one column per word in the order of the sentence. A 1 in row i and column j means: word i adds the row of word j.



	bootet	*kat*	*divin*	*feesh*
bootet				
kat				
divin				
feesh				

S

Write the arrows into a table S, one row and one column per word in the order of the sentence. A 1 in row i and column j means: word i adds the row of word j.

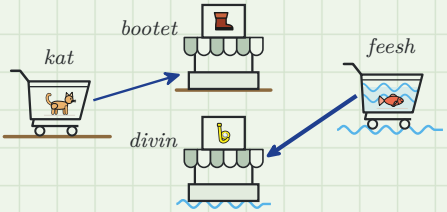


	bootet	*kat*	*divin*	*feesh*
bootet				
kat	1			
divin				
feesh				

S

kat buys at *bootet*: a 1 in row *kat*, column *bootet*.

Write the arrows into a table S, one row and one column per word in the order of the sentence. A 1 in row i and column j means: word i adds the row of word j.



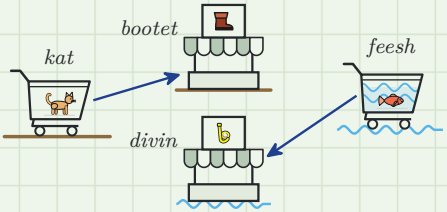
	bootet	*kat*	*divin*	*feesh*
bootet				
kat	1			
divin				
feesh			1	

S

kat buys at *bootet*: a 1 in row *kat*, column *bootet*.

feesh buys at *divin*: a 1 in row *feesh*, column *divin*.

Write the arrows into a table S, one row and one column per word in the order of the sentence. A 1 in row i and column j means: word i adds the row of word j.



	bootet	*kat*	*divin*	*feesh*
bootet	1			
kat	1	1		
divin			1	
feesh			1	1

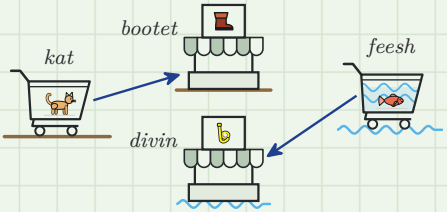
S

kat buys at *bootet*: a 1 in row *kat*, column *bootet*.

feesh buys at *divin*: a 1 in row *feesh*, column *divin*.

Every word keeps its own row: a 1 on the diagonal.

Write the arrows into a table S, one row and one column per word in the order of the sentence. A 1 in row i and column j means: word i adds the row of word j.



	bootet	*kat*	*divin*	*feesh*
bootet	**1**	0	0	0
kat	**1**	**1**	0	0
divin	0	0	**1**	0
feesh	0	0	**1**	**1**

S

kat buys at *bootet*: a 1 in row *kat*, column *bootet*.

feesh buys at *divin*: a 1 in row *feesh*, column *divin*.

Every word keeps its own row: a 1 on the diagonal.

Everywhere else 0: no arrow, nothing to add.

Each row of S marks the rows to add. The product SV adds them up: its row i is the sum of the rows of V that row i of S marks with a 1.

$V = E$
for now: part 4 comes back to V

	LEGS	FINS	WINGS	RUNNING	BOOTS	SWIMMING	DIVING	
	0	0	0	0	**1**	0	0	*bootet*
	1	0	0	0	0	0	0	*kat*
	0	0	0	0	0	0	**1**	*divin*
	0	**1**	0	0	0	0	0	*feesh*

S

	bootet	*kat*	*divin*	*feesh*
bootet	**1**	0	0	0
kat	**1**	**1**	0	0
divin	0	0	**1**	0
feesh	0	0	**1**	**1**

SV

Each row of S marks the rows to add. The product SV adds them up: its row i is the sum of the rows of V that row i of S marks with a 1.

$V = E$
for now: part 4 comes back to V

	LEGS	FINS	WINGS	RUNNING	BOOTS	SWIMMING	DIVING	
	0	0	0	0	**1**	0	0	*bootet*
	1	0	0	0	0	0	0	*kat*
	0	0	0	0	0	0	**1**	*divin*
	0	**1**	0	0	0	0	0	*feesh*

	bootet	*kat*	*divin*	*feesh*
bootet	**1**	0	0	0
kat	**1**	**1**	0	0
divin	0	0	**1**	0
feesh	0	0	**1**	**1**

S

1	0	0	0	**1**	0	0	***pussinbootz***

SV



Each row of S marks the rows to add. The product SV adds them up: its row i is the sum of the rows of V that row i of S marks with a 1.

$V = E$
for now: part 4 comes back to V

	LEGS	FINS	WINGS	RUNNING	BOOTS	SWIMMING	DIVING	
	0	0	0	0	**1**	0	0	*bootet*
	1	0	0	0	0	0	0	*kat*
	0	0	0	0	0	0	**1**	*divin*
	0	**1**	0	0	0	0	0	*feesh*

	bootet	*kat*	*divin*	*feesh*
bootet	**1**	0	0	0
kat	**1**	**1**	0	0
divin	0	0	**1**	0
feesh	0	0	**1**	**1**

S

1	0	0	0	**1**	0	0	***pussinbootz***
0	**1**	0	0	0	0	**1**	***divefeesh***

SV





Each row of S marks the rows to add. The product SV adds them up: its row i is the sum of the rows of V that row i of S marks with a 1.

$V = E$
for now: part 4 comes back to V

	LEGS	FINS	WINGS	RUNNING	BOOTS	SWIMMING	DIVING	
	0	0	0	0	**1**	0	0	*bootet*
	1	0	0	0	0	0	0	*kat*
	0	0	0	0	0	0	**1**	*divin*
	0	**1**	0	0	0	0	0	*feesh*

S

	bootet	*kat*	*divin*	*feesh*
bootet	**1**	0	0	0
kat	**1**	**1**	0	0
divin	0	0	**1**	0
feesh	0	0	**1**	**1**

SV

bootet	0	0	0	0	**1**	0	0	*bootet*
pussinbootz	**1**	0	0	0	**1**	0	0	***pussinbootz***
divin	0	0	0	0	0	0	**1**	*divin*
divefeesh	0	**1**	0	0	0	0	**1**	***divefeesh***





Each row of S marks the rows to add. The product SV adds them up: its row i is the sum of the rows of V that row i of S marks with a 1.

kat is now *pussinbootz*, and *feesh* is *divefeesh*: the picture you drew in the game.

$V = E$
for now: part 4 comes back to V

	LEGS	FINS	WINGS	RUNNING	BOOTS	SWIMMING	DIVING	
	0	0	0	0	**1**	0	0	*bootet*
	1	0	0	0	0	0	0	*kat*
	0	0	0	0	0	0	**1**	*divin*
	0	**1**	0	0	0	0	0	*feesh*

S

	bootet	*kat*	*divin*	*feesh*
bootet	**1**	0	0	0
kat	**1**	**1**	0	0
divin	0	0	**1**	0
feesh	0	0	**1**	**1**

SV

	0	0	0	0	**1**	0	0	*bootet*
	1	0	0	0	**1**	0	0	***pussinbootz***
	0	0	0	0	0	0	**1**	*divin*
	0	**1**	0	0	0	0	**1**	***divefeesh***





Now drop the mall and keep the words where it put them. Every word gets three rows: its value, its query and its key.



bootet

Its **value** is its row of E: what it hands over, as a stand hands over what it sells.

value



kat

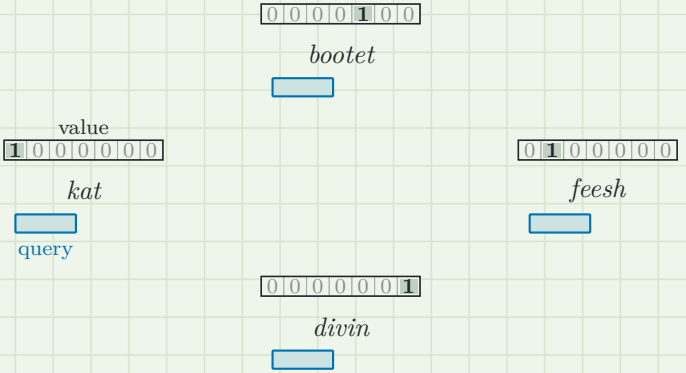


feesh



divin

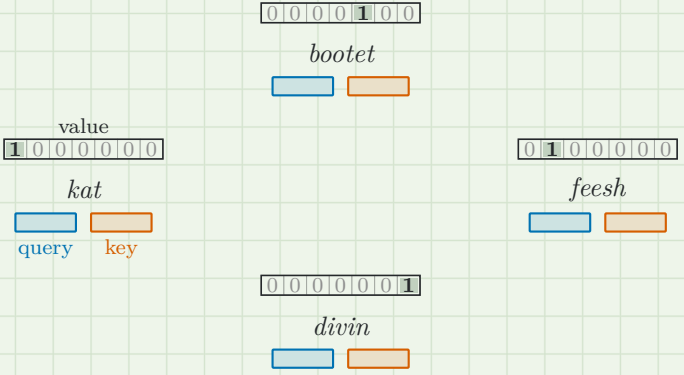
Now drop the mall and keep the words where it put them. Every word gets three rows: its value, its query and its key.



Its **value** is its row of E: what it hands over, as a stand hands over what it sells.

Its **query** says what it looks for, as a shopper does.

Now drop the mall and keep the words where it put them. Every word gets three rows: its value, its query and its key.

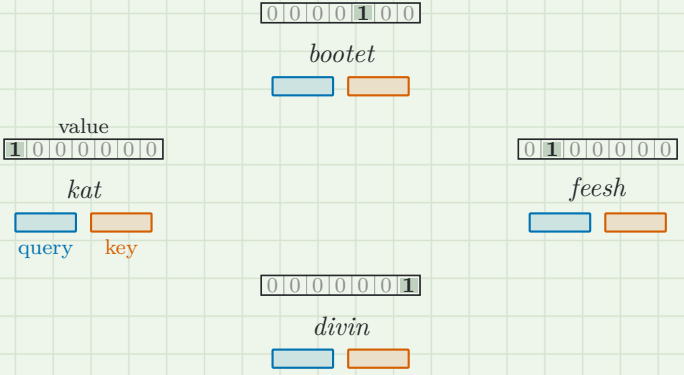


Its **value** is its row of E: what it hands over, as a stand hands over what it sells.

Its **query** says what it looks for, as a shopper does.

Its **key** says what it offers, as the sign of a stand does.

Now drop the mall and keep the words where it put them. Every word gets three rows: its value, its query and its key.



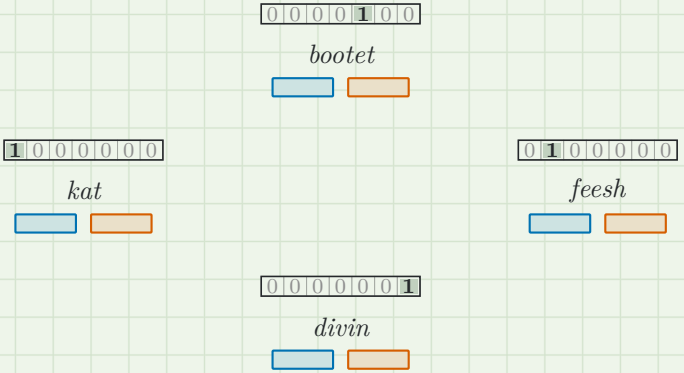
Its **value** is its row of E: what it hands over, as a stand hands over what it sells.

Its **query** says what it looks for, as a shopper does.

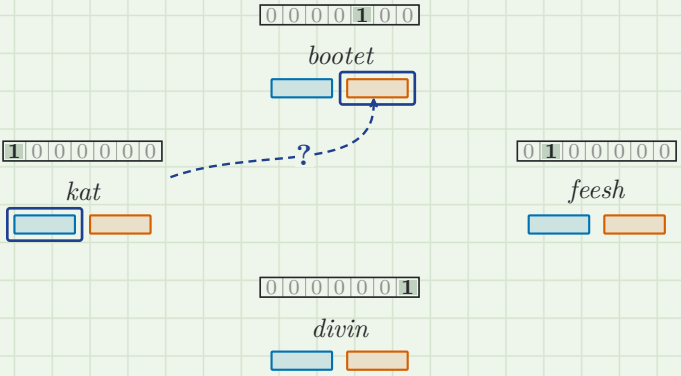
Its **key** says what it offers, as the sign of a stand does.

So every word can shop and sell. Part 4 computes queries and keys from E.

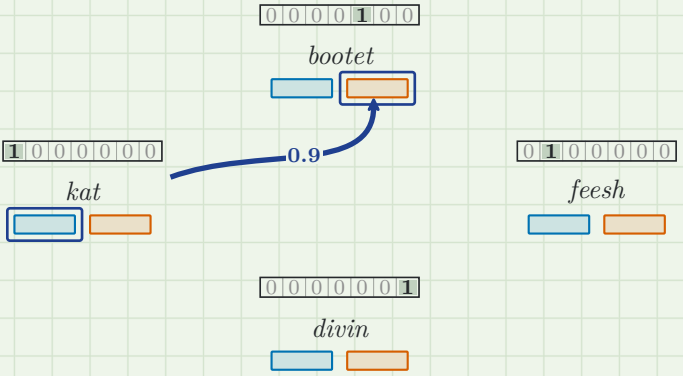
The query of a word checks the key of every word, its own included: their inner product is the strength of the arrow. These numbers are made up.



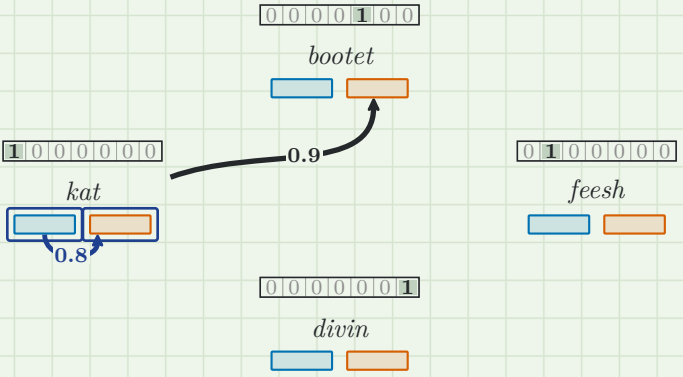
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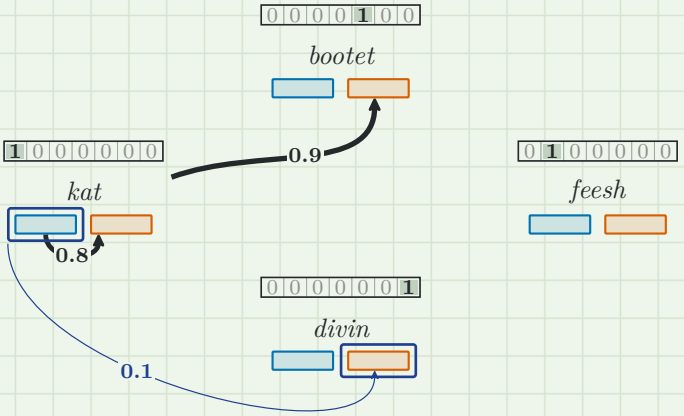
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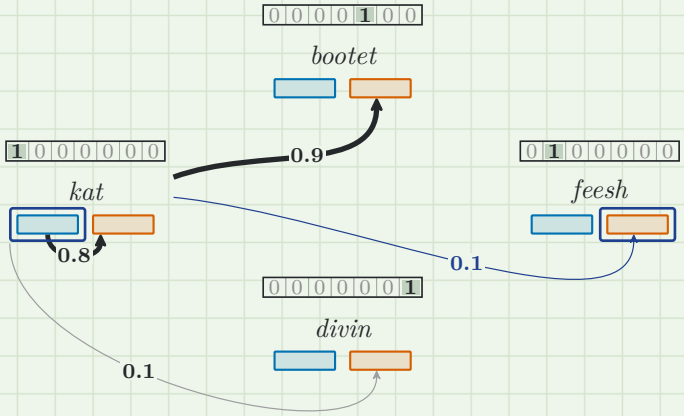
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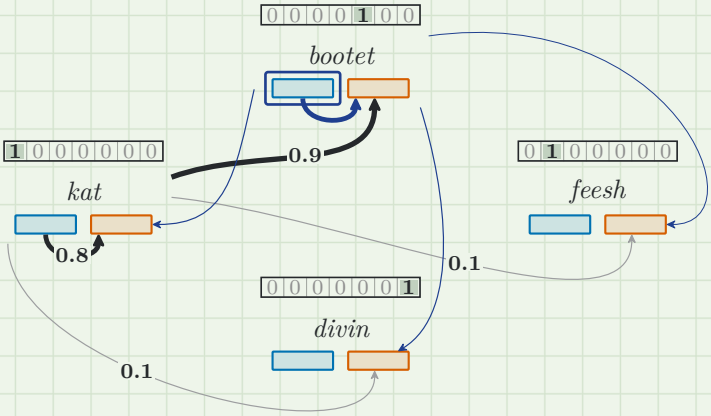
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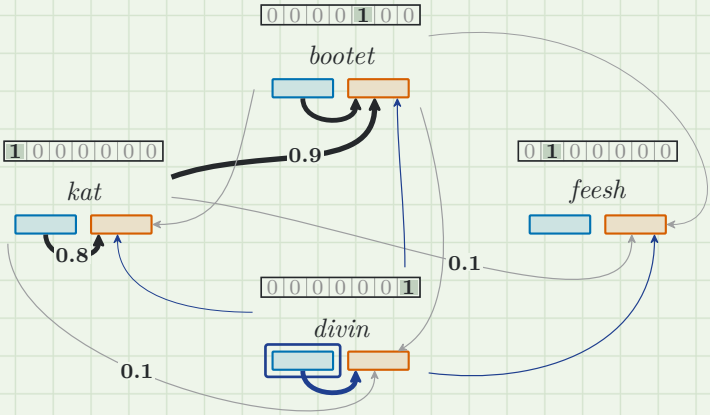
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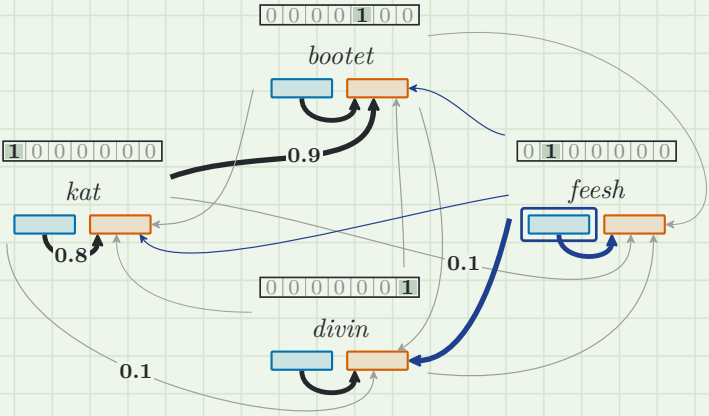
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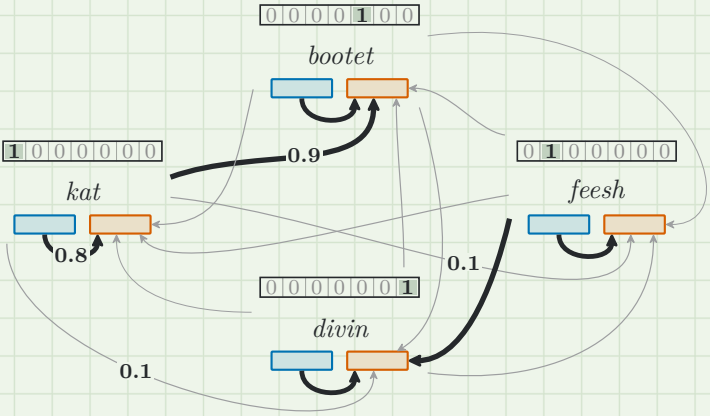
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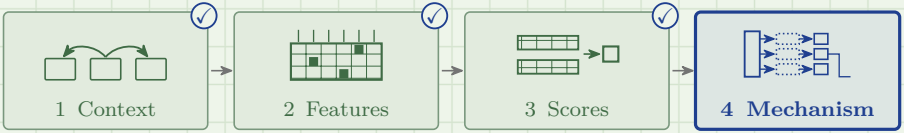
	bootet	*kat*	*divin*	*feesh*
bootet	**0.8**	0.1	0.1	0.0
kat	**0.9**	**0.8**	0.1	0.1
divin	0.1	0.0	**0.8**	0.1
feesh	0.1	0.1	**0.9**	**0.8**

the attention matrix S

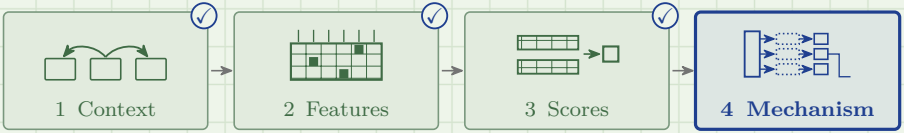
Row i holds the arrows of word i. Its large entries sit where the mall had a 1.

Entry 4

The attention mechanism



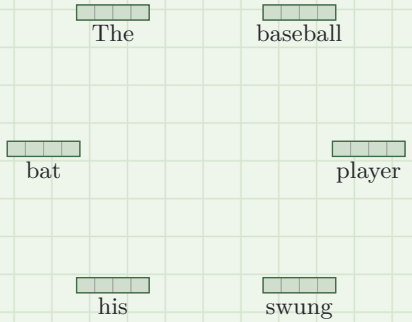
So far. On our sentence, every word found the rows to add: the arrows of the mall make the matrix S, the new rows are SV, and the inner products of queries and keys give the strength of every arrow.



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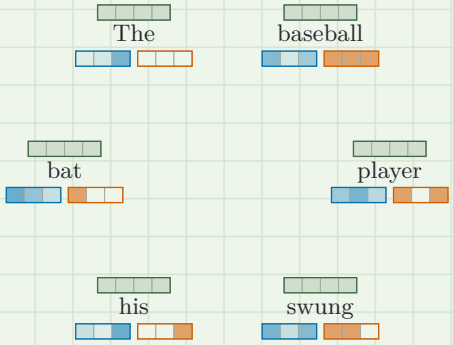
Now. The same recipe for any sentence, in five steps. Then the attention mechanism as blocks, with every size a letter.

Take any sentence, here six words, each with its row: its embedding. Every word turns its row into two short rows, a query and a key. All the numbers here are made up.



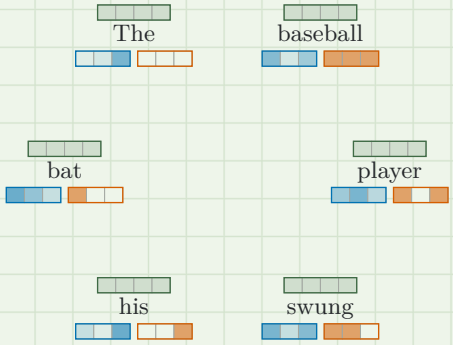
1. **Produce queries and keys**
2. Create the attention graph
3. Softmax the scores into weights
4. Create value vectors
5. Add up the weighted values

Take any sentence, here six words, each with its row: its embedding. Every word turns its row into two short rows, a query and a key. All the numbers here are made up.



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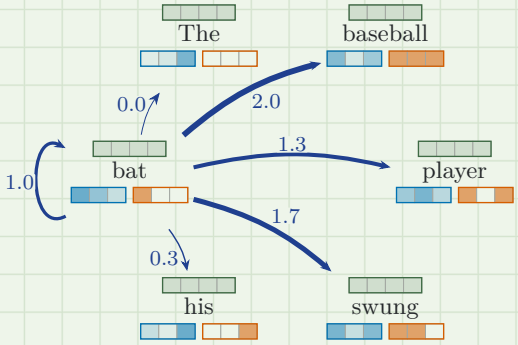
1. **Produce queries and keys**
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The query and the key of **bat**:

query	1.0	0.7	0.3	key	1	0	0

In every row, the darker a cell, the larger its number.

The query of a word meets the key of every word, its own included. Their inner product, the *score*, sets the width of an arrow. All the arrows make the *attention graph*.



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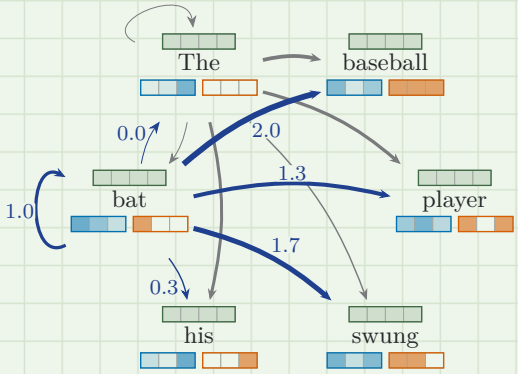
The arrow from **bat** to **baseball**:

query of bat	1.0	0.7	0.3
key of baseball	1	1	1

inner product $1.0 + 0.7 + 0.3 = 2.0$

bat reads **baseball** and **swung** most: here a bat is a club, as at the start of the lecture.

The query of a word meets the key of every word, its own included. Their inner product, the *score*, sets the width of an arrow. All the arrows make the *attention graph*.



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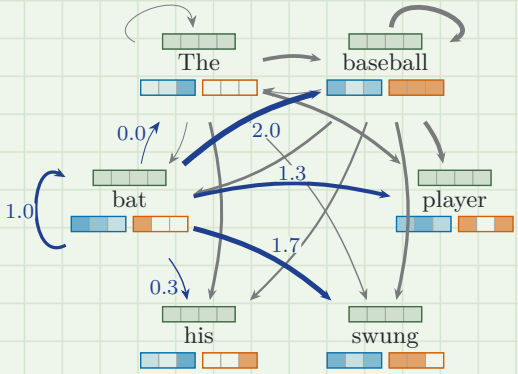
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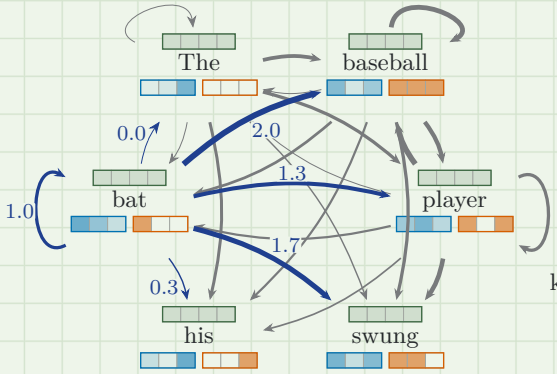
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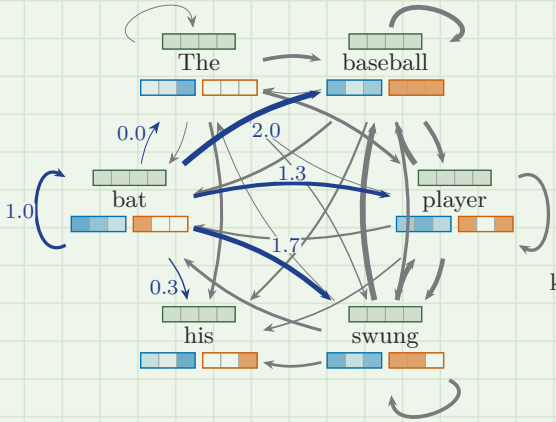
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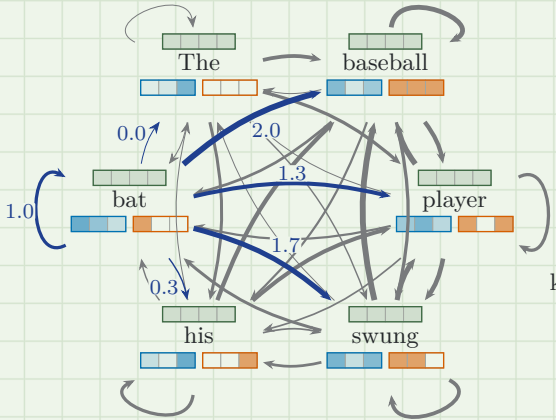
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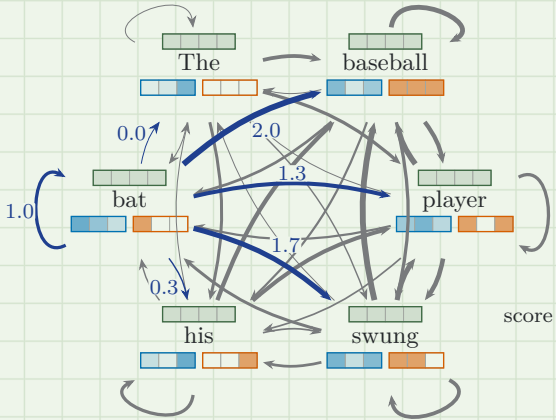
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Scores can be any numbers. The softmax raises e to every score of a word and divides by their total: the weights of the word, all positive and adding up to 1.

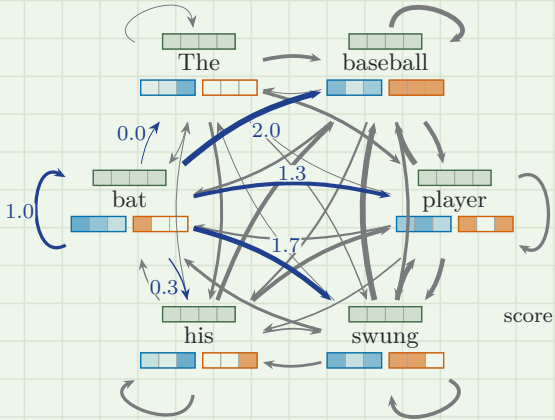


1. Produce queries and keys
2. Create the attention graph
3. **Softmax the scores into weights**
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$$\text{weight} = \frac{e^{\text{score}}}{\text{sum of } e^{\text{score}} \text{ over all words}}$$

	The	baseball	player	swung	his	bat
score of **bat**	0.0	2.0	1.3	1.7	0.3	1.0

Scores can be any numbers. The softmax raises e to every score of a word and divides by their total: the weights of the word, all positive and adding up to 1.

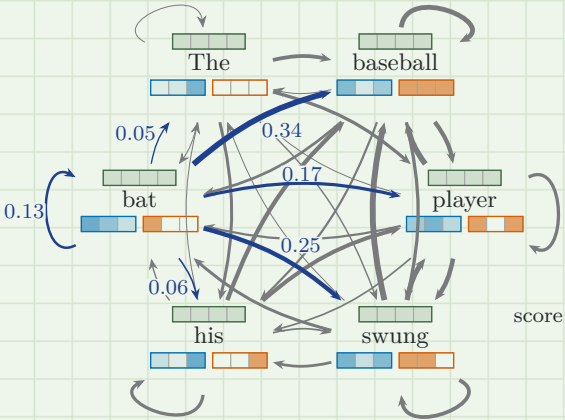


1. Produce queries and keys
2. Create the attention graph
3. **Softmax the scores into weights**
4. Create value vectors
5. Add up the weighted values

$$\text{weight} = \frac{e^{\text{score}}}{\text{sum of } e^{\text{score}} \text{ over all words}}$$

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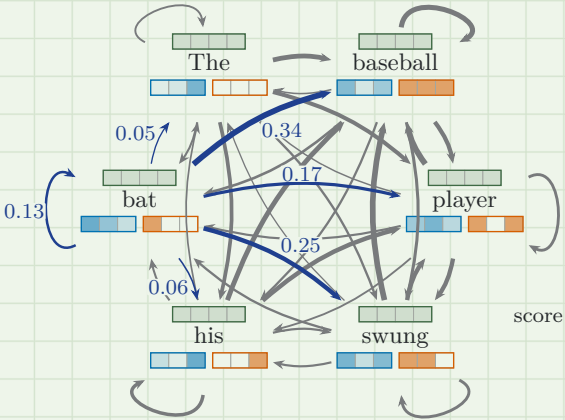
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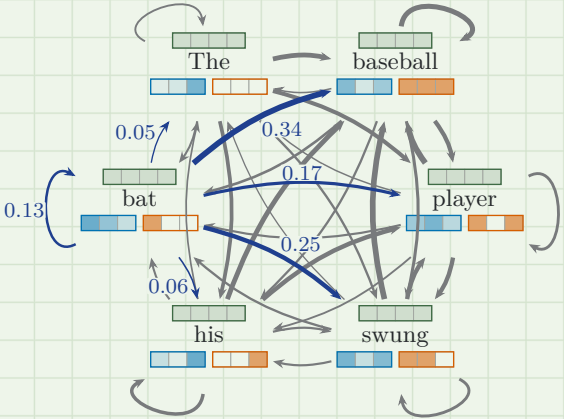
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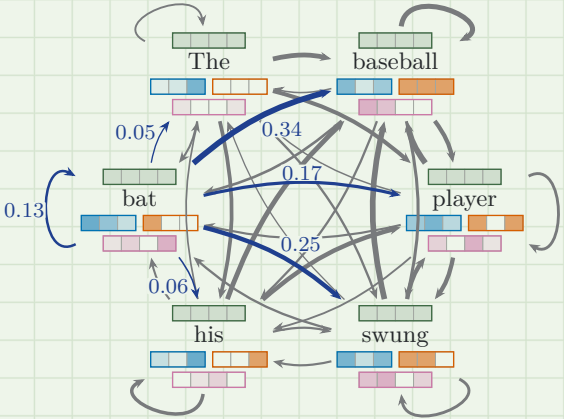
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Every word turns its row into a third short row, its *value*: what the word hands over to every word that reads it.



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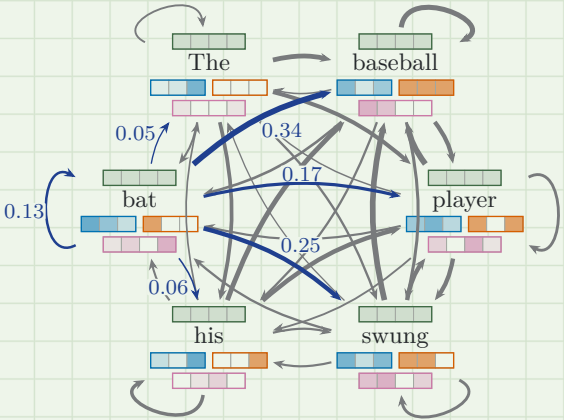
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The value of **bat**:

value	0.25	0.5	0.0	1.0

A value here has four numbers, as many as an embedding.

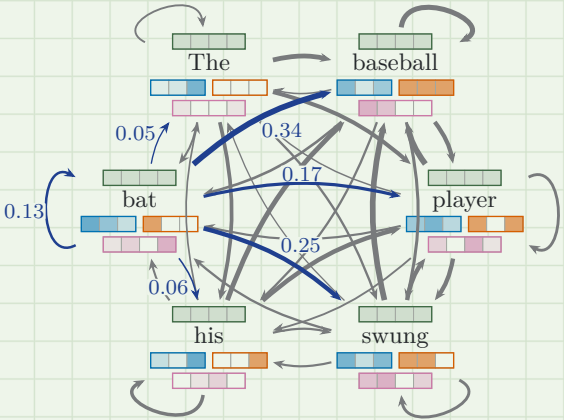
The new row of a word, its new embedding, adds up the values of all the words, each multiplied by the weight of its arrow.



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0.25	0.0	0.0	0.25	The
1.0	0.75	0.0	0.0	baseball
0.5	0.0	1.0	0.25	player
0.75	1.0	0.0	0.5	swung
0.0	0.25	0.5	0.0	his
0.25	0.5	0.0	1.0	bat

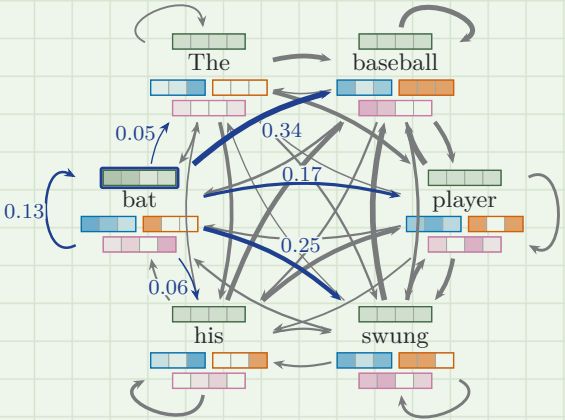
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0.05×	0.25	0.0	0.0	0.25	The
0.34×	1.0	0.75	0.0	0.0	baseball
0.17×	0.5	0.0	1.0	0.25	player
0.25×	0.75	1.0	0.0	0.5	swung
0.06×	0.0	0.25	0.5	0.0	his
0.13×	0.25	0.5	0.0	1.0	bat

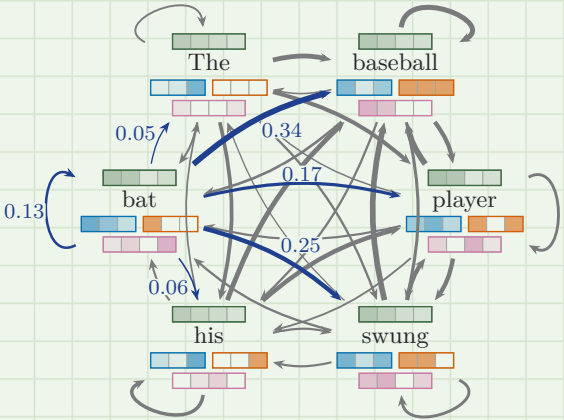
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=	0.7	0.6	0.2	0.3	new row of **bat**

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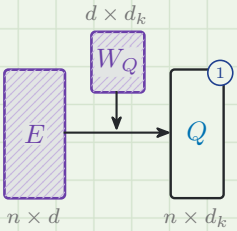
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Every word does the same with its own weights.

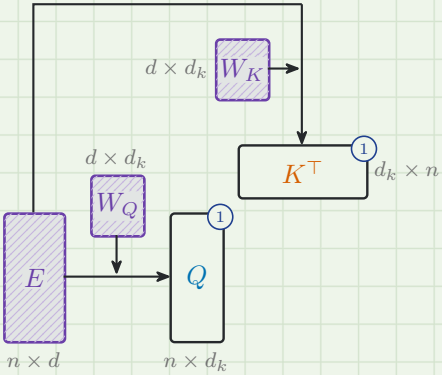
Stack the rows of n words into a block E: the recipe becomes products of matrices and a softmax. Training learns the violet, hatched blocks: E and the three blocks of weights.



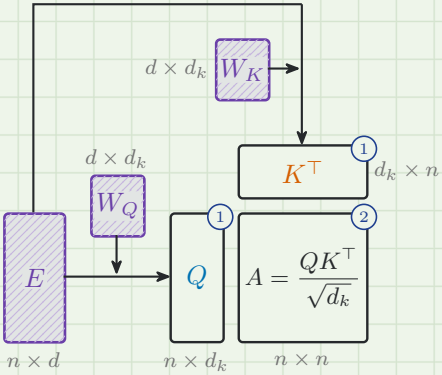
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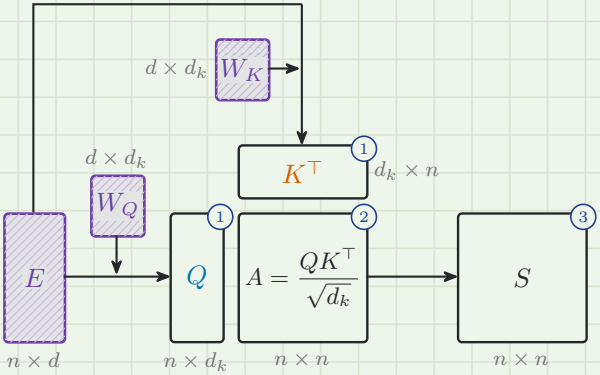


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We divide by $\sqrt{d_k}$: it keeps the scores in a range where the softmax works.

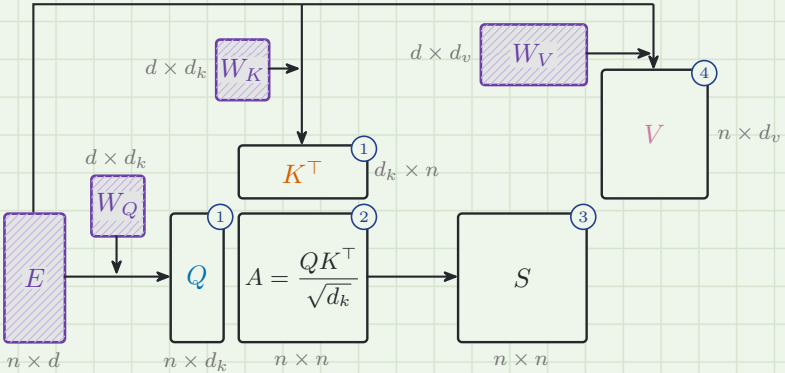
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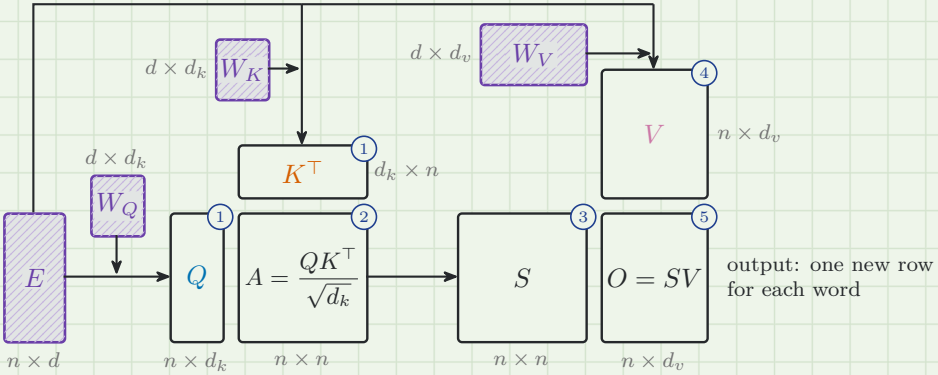
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